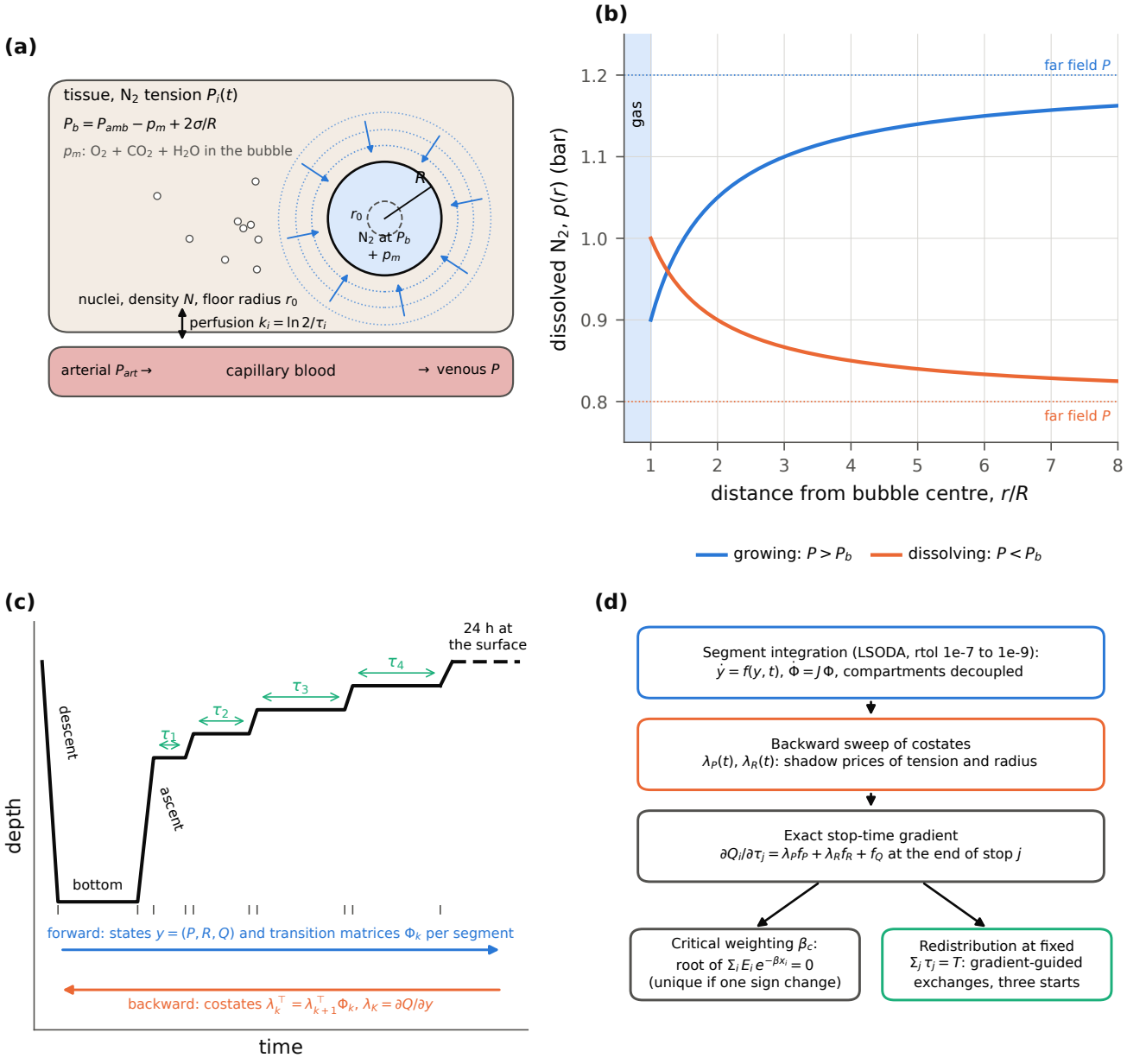
(a)
tissue, N_2 tension $P_i(t)$
$P_b = P_{amb} - p_m + 2\sigma/R$
p_m: O_2 + CO_2 + H_2O in the bubble
R
r_0
N_2 at P_b + p_m
nuclei, density N, floor radius r_0
perfusion $k_i = \ln 2/\tau_i$
arterial P_{art} →
capillary blood
→ venous P
(b)
gas
far field P
far field P
dissolved N_2, $p(r)$ (bar)
distance from bubble centre, r/R
growing: $P > P_b$
dissolving: $P < P_b$
(c)
24 h at the surface
descent
ascent
bottom
τ_1
τ_2
τ_3
τ_4
depth
time
forward: states $y = (P, R, Q)$ and transition matrices Φ_k per segment
backward: costates $\lambda_k^\top = \lambda_{k+1}^\top \Phi_k$, $\lambda_K = \partial Q/\partial y$
(d)
Segment integration (LSODA, rtol 1e-7 to 1e-9):
$\dot{y} = f(y, t)$, $\dot{\Phi} = J\Phi$, compartments decoupled
Backward sweep of costates
$\lambda_P(t)$, $\lambda_R(t)$: shadow prices of tension and radius
Exact stop-time gradient
$\partial Q_l/\partial \tau_j = \lambda_P f_P + \lambda_R f_R + f_Q$ at the end of stop j
Critical weighting β_c:
root of $\Sigma_i E_i e^{-\beta x_i} = 0$
(unique if one sign change)
Redistribution at fixed
$\Sigma_j \tau_j = T$: gradient-guided
exchanges, three starts